



Plumbing AI Compliance Assistant: A Retrieval- Augmented Generation Proof of Concept

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Executive Summary

Foleys staff regularly need quick, reliable answers from New Zealand plumbing and building compliance documents while planning work, quoting, ordering materials, or resolving on-site questions. In practice, those answers are buried across multiple standards and Building Code guidance, and the work often depends on time-consuming manual searches and informal knowledge held by senior staff. This project developed a proof-of-concept Compliance AI Assistant to reduce search time while preserving accountability through evidence-based responses.

The system was built using a Retrieval-Augmented Generation (RAG) approach in which official documents act as the sole source of truth. A fixed corpus of 27 compliance documents was established and processed into chunked, metadata-rich records with page-level traceability. The final ingestion workflow uses fast page-based extraction, conservative text normalisation, and token-aware chunking. Each chunk is indexed in Pinecone using two complementary indexes: a dense semantic index and a sparse lexical index. At query time, the system retrieves 20 candidates from each index, merges and deduplicates results, then applies Pinecone reranking to select the top five evidence passages for answer generation using gpt-4.1-mini.

Evaluation using representative questions provided by Foleys showed stable retrieval behavior and generally strong answer quality when the relevant clause or table was retrieved. The assistant produced structured outputs that separate AS/NZS requirements from New Zealand Building Code (NZBC) material and includes exact evidence quotes with page-level S3 links in the form, enabling direct verification. When evidence was insufficient, the system stated limitations rather than guessing. End-to-end latency was acceptable for proof-of-concept, with generation time contributing most variability. Overall, the prototype demonstrates that an evidence-grounded RAG assistant can act as a practical navigation layer over compliance documents, while highlighting the need for continued work on retrieval coverage, ambiguity handling, and larger-scale validation.

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1. Introduction

Compliance work in the building and plumbing industry is not only technical, it is also time sensitive. In Foleys' context, staff may need to confirm requirements while planning jobs, quoting, ordering materials, or deciding what is acceptable on-site. These questions are rarely phrased in the same way standards are written. People ask in everyday language, they shorten terms, they use trade shorthand, and they often start with a partial idea rather than a clean specification.

At the same time, New Zealand building regulations and supporting compliance documents are complex, and they are designed to be interpreted within a compliance framework (Ministry of Business, Innovation & Employment, 2014). Building regulations are performance-based, so they describe what must be achieved rather than prescribing one exact solution. In practice, people rely on compliance pathways like Acceptable Solutions and Verification Methods, or they propose Alternative Solutions with supporting evidence. For plumbing and drainage, the AS/NZS 3500 standards form a key pathway for "how to comply" under commonly used Acceptable Solutions (Ministry of Business, Innovation & Employment, 2026). That combination is powerful, but it also creates an information problem: the relevant answer is often present, but it is buried inside a large set of documents, and the user needs to find it quickly and interpret it correctly.

This project sets out to design a Retrieval-Augmented Generation assistant that makes compliance information easier to access while preserving accountability. The goal was not to replace professional judgement or regulatory responsibility. It was to reduce time spent searching, improve consistency in referencing authoritative documents, and create a more transparent way to answer common questions, including links and citations so users can verify the source.

2. Problem Statement

From discussions with the project context and typical operational realities, the core problem can be summarised as follows:

- **Manual searching is slow and disruptive.** Locating answers within technical standards often involves repeated keyword searches, extensive scrolling, and cross-referencing across multiple PDF documents.
- **Regulations are hard to navigate.** Standards and Acceptable Solutions are drafted for accuracy, not for quick search by time-pressed staff.

- **Compliance risk carries real cost.** Misinterpretation or overlooking a requirement can cause rework, delays, failed inspections, penalties, and serious long-term reputational damage.
- **Knowledge can be concentrated in senior staff.** When answers depend on a small group of experienced people, it creates a bottleneck and increases the risk of inconsistent advice when those staff are unavailable.
- **Version changes add uncertainty.** Differences between editions (for example, 2022 versus 2025 editions) can create confusion, especially when newer editions are not yet referenced by legislation but may still be used as Alternative Solutions.

This is fundamentally a data and decision-support problem: how can we retrieve the right evidence from authoritative documents quickly, then present it in a way that users trust and can audit.

3. Project Aims and Scope

3.1 Aims

The project aimed to build and test a proof-of-concept that can:

1. **Standards Retrieval:** Allow users to ask natural language questions and receive accurate, up-to-date answers grounded in official compliance documents.
2. **Contextual Understanding:** Handle paraphrasing and variations in how questions are asked, and request clarification when queries are ambiguous.
3. **Document Navigation:** Provide citations and deep links to relevant sections, and summarise long sections into concise, actionable points.
4. **Multi-format Support:** Return text answers and optionally provide downloadable references (such as PDF links or supporting diagrams, when available).
5. **Updated Standards Awareness:** Summarise changes between document versions and note the compliance status (for example, when an edition is not yet referenced by the Building Code but could be used as an Alternative Solution).
6. **Feedback Loop:** Enable users to rate responses or flag inaccuracies, supporting iterative improvement.

4. Background and Standards Context

4.1 Performance-based regulation and compliance pathways

New Zealand building regulations operate on a performance basis. This means the law describes outcomes that must be achieved (for example, water systems must be safe and hygienic), rather than prescribing a single method. Compliance pathways support how those outcomes are shown in practice:

- Acceptable Solutions: Prescriptive paths that, if followed, are accepted as compliant.
- Verification Methods: Methods based on testing or calculations that are used to show compliance.
- Alternative Solutions: Different approaches that can be put forward and justified using supporting evidence.

In day-to-day plumbing practice, AS/NZS standards are relied on as a practical guide for applying requirements in real situations. The AS/NZS 3500 series is commonly regarded as the primary reference for plumbing and drainage work (Ministry of Business, Innovation & Employment, 2021). Alongside this, Building Code linkage documents and Acceptable Solutions, including those related to the G12 Water Supplies area, are regularly used within the industry and help bring structure to what would otherwise be a wide and technically demanding set of standards (Ministry of Business, Innovation & Employment, 2026).

4.2 Standards versions and practical interpretation

A practical issue arises from the fact that multiple editions of the same standard can exist, while only certain versions are referenced within Acceptable Solutions at any given time (Ministry of Business, Innovation & Employment, 2026). Within the context of this project:

- The 2021/2022 editions were treated as the current Acceptable Solutions reference point (Ministry of Business, Innovation & Employment, 2014).
- 2025 editions were noted as not yet referenced by legislation but could still be used as Alternative Solutions.
- Older editions (for example 2018 versions in some cases) were considered superseded and not recommended as the main compliance reference.

This matters for a RAG assistant because the system must not only retrieve text, it must also communicate the status and context of that text.

5. Method

5.1 Overall approach

This project followed a data-centric approach to the development of a proof-of-concept Compliance AI Assistant for New Zealand building and plumbing standards. Rather than aiming for open-ended conversation, the emphasis was placed on supporting compliance-related decision making by ensuring accurate information retrieval and the clear, transparent presentation of authoritative sources.

The system was designed using a RAG paradigm, where official standards documents form the sole source of truth. Large Language Models (LLM) are used only to synthesise and summarise retrieved content, rather than to generate answers independently. This ensures that responses remain grounded in verifiable evidence.

The work followed a three-stage process. The process started by defining a fixed collection of official compliance documents and converting them into a searchable format, while still maintaining page-level traceability. The next stage then focused on developing a hybrid retrieval strategy that combined semantic search with lexical matching, enabling the system to respond to both everyday language questions and more specialised technical terminology. In the final stage, a controlled generation layer was used to generate structured responses, with each answer clearly supported by citations that pointed back to the original source documents.

Overall, this approach mirrors the way compliance tasks are carried out in real settings, where clarity, accuracy, and traceability matter, and it treats the system as a tool to support decision making rather than something intended to replace professional judgement.

5.2 System Architecture Overview

The system is built using a multi-tier architecture that separates the user interface, application logic, and data storage. A Next.js frontend is used to deliver the chat interface and admin dashboard, while a FastAPI backend runs the RAG pipeline through a combination of WebSockets and REST Application Programming Interface (API). Information retrieval is managed through Pinecone hybrid search, using both a dense index and a sparse index with reranking, and OpenAI gpt-4.1-mini is used to generate the final responses with evidence-based

citations. PostgreSQL is used to store user data and structured analytics, while MongoDB Atlas holds conversation histories, messages, and feedback. PDF documents are uploaded to AWS S3, and citations are linked to the exact source pages.

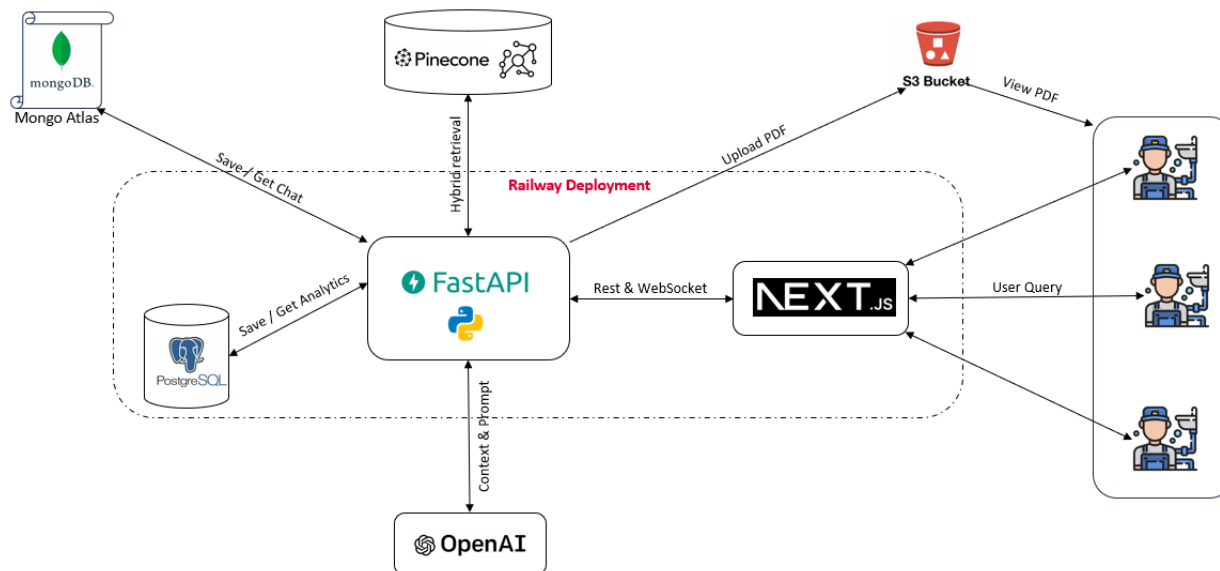


Figure 1- System Architecture

5.3 Database Design and Data Storage

The system uses a hybrid data storage approach to accommodate both structured operational data and more flexible conversational data. Relational data storage is applied in areas where consistency, data integrity, and analytical querying are important, while document-based storage is used to manage conversational context and message-level metadata. A separate vector database is employed to enable effective retrieval across the compliance document corpus.

5.3.1 PostgreSQL (Relational Storage)

PostgreSQL is used to store user accounts and structured analytics data. This database underpins core system functions such as authentication, role-based access control, administrative reporting, and overall system monitoring. The data held in PostgreSQL follow a relatively stable structure and are well suited to relational constraints and indexing, which supports consistency and efficient querying. Postgres schema see in [appendix](#).

5.3.2 MongoDB Atlas (Document Storage)

MongoDB Atlas is used to store conversational data, which is naturally hierarchical and benefits from a flexible schema (MongoDB, n.d.). This database captures the full chat history, message sequencing, and detailed metadata associated with each assistant response. MongoDB schema see in [appendix](#).

5.3.3 Pinecone (Vector Database for Retrieval)

Pinecone is used to store and retrieve document chunks for the retrieval-augmented generation pipeline. Two separate indexes are maintained to enable hybrid retrieval (Pinecone Systems, Inc, n.d.). Pinecone metadata schema see in [appendix](#).

5.4 Document Ingestion Pipeline

The document ingestion pipeline transforms compliance PDFs into a structured and searchable representation that supports retrieval-augmented generation while preserving traceability to original source documents. The pipeline was designed with a strong emphasis on regulatory transparency, reproducibility, and alignment with LLM constraints.

5.4.1 Document Upload and Source Management

All compliance documents are uploaded manually to an AWS S3 bucket and handled as immutable source artefacts. This setup ensures that the system works with a fixed and auditable set of documents, which is particularly important in a compliance setting. AWS S3 was chosen because it offers stable, publicly accessible URLs that can be shared directly with end users.

Rather than generating page-specific links during ingestion, the pipeline stores the base S3 document link together with the page number as metadata. During answer generation, these values are combined to produce citations in the format `<s3-link>#page=<page_number>`, allowing users to open the exact supporting page in the original PDF (Adobe, 2007).

5.4.2 PDF Text Extraction

Text extraction is performed at the page level to ensure that all retrieved evidence can be traced back to an exact location in the source document. Early versions of the pipeline used pdfplumber because of its accuracy on complex PDF layouts. When processing large standards documents, this approach proved to be computationally slow and unsuitable for repeated ingestion runs.

The final implementation uses PyMuPDF, which provides significantly faster page-level text extraction while still preserving page boundaries (PyMuPDF, n.d.). Each page is processed one at a time, with the extracted text explicitly linked to its corresponding page number. Maintaining this link is important because the page numbers are later shown directly to users as part of the citation process.

5.4.3 Text Cleaning and Normalisation

Extracted text is lightly cleaned to reduce noise introduced by PDF formatting. This step includes normalising whitespace, collapsing excessive line breaks, and removing layout artefacts that could interfere with tokenisation or embedding. Cleaning is intentionally conservative to avoid altering regulatory wording, as even minor textual changes could affect compliance interpretation.

The purpose of this step is to improve consistency across documents with different layouts rather than to rewrite or summarise content.

5.4.4 Metadata Construction and Version Context

Each document is assigned structured metadata before chunking. Metadata fields include document title, filename, standard type, edition year, and currency status. These attributes are inferred based on filename conventions and recognised naming patterns commonly used in New Zealand compliance documents.

When multiple editions of a standard are available, the metadata records whether a document aligns with a currently referenced Building Code edition or represents a newer version that may only be applicable as an Alternative Solution. Storing this information explicitly provides important regulatory context during retrieval and reduces the risk of users misinterpreting answers.

5.4.4 Token-Aware Chunking

Chunking is carried out using a token-aware strategy that is aligned with the tokenisation scheme of the OpenAI language model used for answer generation. The system applies the `cl100k_base` token encoding (OpenAI, n.d.), which is consistent with the token units used by OpenAI models when determining context length and token limits (OpenAI, 2022). As a result, the chunk sizes defined during ingestion directly reflect how the language model later interprets and processes text at inference time.

A token-based approach was chosen instead of character-based chunking to avoid mismatches between preprocessing and generation. Character counts are a poor proxy for actual model context usage, particularly in regulatory documents that include symbols, tables, and technical formatting. By working in token space, the pipeline is able to keep each chunk within a predictable and safe portion of the model's effective context window (Maryamah, et al., 2024).

As part of the ingestion stage, chunks are created to be roughly 1,740 tokens in length, with a 205-token overlap between neighboring chunks. These parameters were set to manage the trade-off between retrieving enough relevant information and keeping processing costs reasonable. Setting the chunk size to 1,740 tokens allows related regulatory content, such as full clauses or explanations linked to tables, to remain grouped together without creating chunks that are too large for reliable semantic matching. Where content extends across chunk boundaries, a 205-token overlap is introduced between neighbouring chunks. This approach helps preserve meaning across splits and lowers the risk that the retrieval step returns only partial or disconnected evidence.

5.4.5 Vector Indexing and Upload

After chunking and metadata construction are completed, all document segments are indexed in Pinecone, which functions as the system's vector database for retrieval. Pinecone was chosen because it supports scalable vector search, hybrid retrieval approaches, and metadata-aware querying, all of which are important when working with large collections of regulatory documents.

To support different but complementary retrieval behaviors, the system maintains two separate Pinecone indexes: one dense semantic index and one sparse lexical index. Both indexes are deployed on AWS in the us-east-1 region and run in on-demand capacity mode, which allows the system to scale automatically with query volume without the need for manual capacity planning.

- **Dense Vector Index**

The dense index is intended to model semantic similarity between user queries and individual document chunks. Each chunk is embedded using the **llama-text-embed-v2** model, which generates fixed-length vector representations with a dimensionality of 1024. This makes it possible for the system to retrieve conceptually relevant sections even when the user's phrasing differs from the language used in the standards documents.

In addition to the embeddings themselves, the dense index also stores the relevant metadata for each chunk. This metadata includes the document title, filename, standard type, edition year, currency status, page number, and the base S3 file link. Keeping metadata together with the embeddings makes it possible to apply filters and generate citations later without needing extra lookup steps.

- **Sparse Vector Index**

In parallel, all chunks are indexed in a sparse Pinecone index using the **pinecone-sparse-english-v0** model. This index supports lexical-style retrieval based on term overlap and exact keyword matching. Unlike traditional BM25 implementations, this approach uses Pinecone's managed sparse embedding model, allowing sparse and dense retrieval to be handled consistently within the same infrastructure (Pinecone Systems, Inc, n.d.).

The sparse index stores identical metadata fields to the dense index. This symmetry allows results from both retrieval modes to be merged and compared directly, while still maintaining consistent document references and citation information.

5.4.6 Upload Process and Metadata Consistency

During ingestion, each chunk is uploaded to both the dense and sparse indexes using a shared namespace. Chunk identifiers are generated deterministically to avoid duplication and to maintain a stable mapping between chunks across indexes. Uploading the same chunk content and metadata to both indexes ensures that semantic and lexical retrieval operate over the same underlying evidence base.

Crucially, each indexed record includes both the base S3 document link and the page number from which the chunk was extracted. These fields are not combined during indexing. Rather than being joined earlier in the pipeline, they are exposed independently during retrieval and brought together only at the answer generation stage to form page-level citation links in the format <s3-link>#page=<page_number>. This approach keeps the system flexible while still allowing citations to remain precise and easy to verify.

5.4.6 Evidence Preservation and Traceability

From early on in development, traceability was treated as something the ingestion pipeline had to support rather than an optional feature. Each chunk indexed in Pinecone keeps track of the

document it was taken from, along with the relevant page number. When an answer is produced, this information is passed through to the user as citations, so it is always possible to check the response against the exact page in the original PDF.

Through the combination of page-level extraction, token-aware chunking, and structured metadata, the ingestion pipeline ensures that the assistant operates as an evidence-grounded navigation layer over official compliance documents, rather than as an opaque text generation system.

5.5 RAG Pipeline Implementation

The retrieval-augmented generation (RAG) pipeline integrates hybrid document retrieval with controlled answer generation to provide accurate, evidence-grounded responses to user queries about New Zealand building and plumbing standards. The design prioritises relevance, transparency, and traceability, which are essential in a compliance-focused domain where users must be able to verify answers against official source documents.

5.5.1 Hybrid Retrieval Strategy

When a user submits a query, the system applies a hybrid retrieval process by querying both a dense semantic index and a sparse lexical index in parallel. Both indexes are hosted on Pinecone and deployed on AWS in the **us-east-1** region under a shared namespace, building-standards.

The dense index (**foleys3-rag-dense-v3**) is built on semantic embeddings produced by the **llama-text-embed-v2** model, using a vector dimensionality of 1024. Its role is to retrieve passages that are conceptually related to a user's query, which helps the system respond to natural language questions even when the wording does not closely match the language used in the regulatory documents. At the same time, the sparse index (**foleys3-rag-sparse-v3**) uses Pinecone's managed sparse retrieval model, **pinecone-sparse-english-v0**, to identify passages with strong lexical overlap between the query and the source text (Pinecone Systems, Inc, n.d.). This is particularly important for compliance queries that rely on precise technical terminology (Pinecone Systems, Inc, n.d.).

From each index, the system retrieves an initial set of 20 candidate chunks, resulting in a combined pool of up to 40 candidates before merging. The results from both retrieval streams are merged into a single candidate pool by deduplicating on chunk identifiers. If a chunk is returned

by both the dense and sparse retrieval processes, the higher of the two relevance scores is kept. This merging approach is used to keep potentially useful evidence in the results while avoiding repeated appearances of the same passage in the candidate set.

To limit computational overhead during reranking, the combined pool of candidates is restricted to a maximum of 100 chunks. This constraint is intended to balance retrieval recall with efficiency, ensuring that reranking is carried out on the most relevant candidates instead of the entire retrieved set.

5.5.2 Reranking for Relevance

To improve ranking accuracy beyond raw vector similarity, the capped set of candidate chunks is passed through Pinecone's inference-based reranking model, **bge-reranker-v2-m3**. This reranker assesses the relevance of each retrieved passage against the full user query, rather than depending only on embedding similarity scores (Pinecone Systems, Inc, n.d.).

Reranking is applied to a maximum of 100 candidates, after which only the top 5 reranked chunks are kept for answer generation. Adding this extra ranking stage leads to noticeably better retrieval quality, particularly for complex or multi-part regulatory questions where semantic similarity on its own may return passages that only partially address what is being asked.

5.5.3 Context Construction for Generation

The top 5 reranked passages are assembled into a structured context that is provided to the language model. Each passage is accompanied by its associated metadata, including document title, filename, edition year, currency status, page number, and the base S3 file link. These metadata fields are deliberately retained so they can be exposed directly to the user as citations in the generated answer.

As part of the context construction step, a page-level citation link is formed by combining the stored S3 document URL with the corresponding page number, following the format **<file_link>#page=<page_number>**. This allows users to navigate directly to the relevant page in the original PDF for verification.

The system prompt establishes the assistant's role as a domain expert on New Zealand building and plumbing standards and instructs the model to prioritise accuracy, avoid speculation, and rely

strictly on the provided context. The user prompt defines a strict response structure that separates primary requirements derived from AS/NZS standards from secondary or supplementary requirements derived from New Zealand Building Code or linkage documents. This structured prompting ensures consistency across responses and helps users distinguish between different compliance pathways.

5.5.4 Answer Generation and Citation

Answer generation is performed using the OpenAI gpt-4.1-mini model. This model was selected as a balanced trade-off between reasoning capability, response reliability, and operational cost. In a compliance-focused setting, the primary requirement is faithful summarisation of retrieved evidence rather than open-ended creativity. Empirical testing showed that gpt-4.1-mini consistently followed structured prompts, respected evidence boundaries, and produced stable outputs at lower latency and cost compared to larger general-purpose models, making it suitable for a production-oriented proof-of-concept.

The model is instructed to summarise requirements using bullet points and to include exact quoted evidence copied directly from the retrieved passages. Each quoted passage is required to include the filename, page number, and a clickable citation link formed from the S3 document URL and the corresponding page number. This design makes it possible for users to check the information for themselves, which helps build confidence in the system's outputs.

As a result, each response includes both a structured natural language explanation and a clear set of source references, so that every claim can be traced back to an official standards document.

5.5.5 Design Rationale

The overall RAG pipeline represents a considered balance between retrieving a broad range of relevant information and producing precise, well-supported answers. Hybrid retrieval across dense and sparse indexes improves recall for different types of queries, reranking then refines relevance using a query-aware inference model, and structured prompting is used to maintain clarity, consistency, and traceability in the final output. Taken together, these elements ensure that the assistant is not intended to function as a general conversational system, but rather as a controlled, evidence-based interface for working through complex regulatory documentation.

5.6 User Interaction Flow

The user interaction flow outlines how users submit queries, receive responses supported by evidence, and provide feedback within the compliance assistant. The flow is designed to enable real-time interaction, clear and transparent citation, and the persistent storage of interaction data so it can be analysed at a later stage.

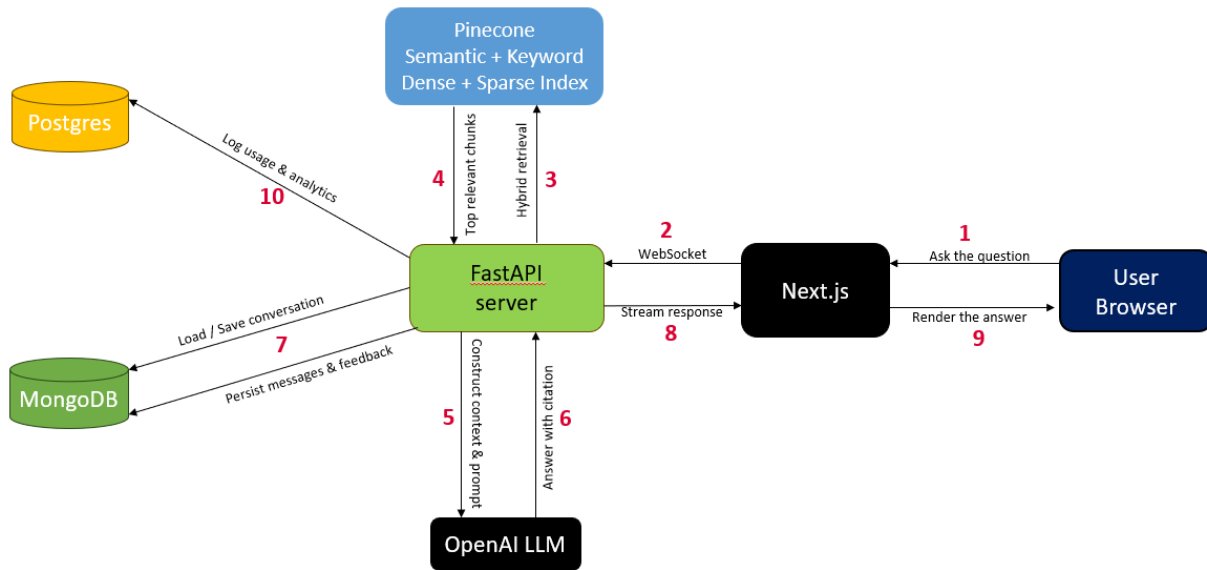


Figure 2 - User Query Flow

5.6.1 User Authentication and Session Management

Authentication in the system is based on JSON Web Tokens. When a user logs in successfully, a signed JWT is generated and stored in an httpOnly cookie. This cookie is then sent automatically with all later API and WebSocket requests. As a result, the backend can link incoming requests to an authenticated user, while the token itself remains inaccessible to client-side scripts.

5.6.2 Real-Time Chat Communication

Real-time interaction for users is achieved via a WebSocket connection linking the Next.js frontend with the FastAPI backend. Using this channel enables low-latency, two-way communication and allows partial responses to be sent back as they are produced, rather than holding everything until the full response is ready. This behaviour becomes particularly noticeable when retrieval and reranking increase processing time, as streaming feedback helps the system feel more responsive to the user.

Once the WebSocket connection has been established, the frontend transmits user queries as structured messages that include the conversation identifier along with the query content. The backend then validates the user session and forwards the request into the retrieval-augmented generation pipeline for further processing.

5.6.3 Query Processing and RAG Invocation

Each time a query is submitted, it initiates a full run of the RAG pipeline. On the backend, hybrid retrieval is carried out by querying the dense semantic Pinecone index (**foleys3-rag-dense-v3**) and the sparse lexical Pinecone index (**foleys3-rag-sparse-v3**) in parallel. From each of these indexes, 20 candidate chunks are returned and combined into a single pool, which is then limited to a maximum of 100 candidates before the reranking step is applied.

The merged candidates are reranked using Pinecone's inference-based reranker, **bge-reranker-v2-m3**, and the top 5 passages are selected for generation. These passages, together with their metadata (filename, edition year, currency status, page number, and S3 file link), are assembled into a structured context for the language model.

5.6.4 Answer Generation and Streaming Response

Answer generation is carried out using an OpenAI gpt-4.1-mini model configured with a low temperature of 0.3 and a maximum output length of 1,500 tokens. The model is provided with a strict system prompt together with the user prompt, which together enforce evidence-based responses, a structured output format, and a clear separation between primary AS/NZS requirements and secondary NZBC or linkage material.

While the response is being generated, output tokens are streamed incrementally to the frontend over the WebSocket connection, rather than being returned only after generation is complete. This allows users to see the answer as it is produced rather than waiting for the full response. The final rendered answer includes formatted sections, bullet points, and exact quoted evidence.

5.6.5 Citation Handling and Source Navigation

For every piece of evidence shown in a generated answer, a clickable citation link is included next to the quoted text. On the backend, this link is formed by starting with the base S3 URL of the source document and then appending the appropriate page number using a URL fragment in the format `<file_link>#page=<page_number>`. Clicking the link takes the user straight from the chat interface to the exact page in the original PDF where the evidence appears.

This page-level linking was treated as an important design decision rather than a cosmetic feature. It allows users to verify claims directly against the source material and helps reinforce confidence in the answers produced by the system.

5.6.6 Conversation and Message Persistence

All conversations and messages are persisted in MongoDB Atlas, which is used to store semi-structured and time-series interaction data. Each conversation record includes metadata such as conversation title, summary, message count, user identifier, and timestamps. Individual message records store the role (user or assistant), message content, sequence number, and additional metadata including token usage, response latency, and the identifiers of retrieved source chunks.

Storing this information enables later analysis of system performance, user behaviour, and retrieval quality without requiring reprocessing of raw logs.

5.6.7 Feedback Collection and Storage

Once an answer has been returned, users are able to provide feedback directly through the chat interface. This feedback includes a categorical rating, for example positive or negative, and may also include optional free-text comments. All feedback is stored in MongoDB Atlas and linked to the associated message and conversation in the system.

Having this feedback mechanism in place allows for a qualitative assessment of answer accuracy, relevance, and clarity. It also provides a useful basis for future improvements to the system, including adjustments to retrieval behaviour or refinements to the prompting strategy.

6. Data

6.1 Data Sources and Composition

The dataset used for this project is made up of 27 official New Zealand documents related to building, plumbing, gas, and regulatory requirements, most of which are provided in PDF format. The collection includes core AS/NZS standards, Building Code Acceptable Solutions and Verification Methods, along with relevant legislative acts and regulations that are commonly referenced when making compliance decisions in the construction and plumbing domain.

The corpus represents a fixed and authoritative source of information for the system. No external or unofficial documents were included, ensuring that all retrieved and generated responses are grounded exclusively in recognised regulatory material.

In total, the dataset contains:

- 3,843 pages
- 8,379,772 characters of text
- 1,616 tables
- 176 images

On average, each document contains approximately 142 pages and 310,362 characters, highlighting the substantial length and complexity of the source material. This scale presents challenges for manual lookup and motivates the use of automated retrieval techniques.

6.2 Document Diversity

The documents vary widely in terms of both their size and overall structure. Some of the larger standards, such as AS/NZS 5601.1:2022 Gas Installations, run to more than 300 pages, whereas smaller Acceptable Solution documents are often fewer than 50 pages. Many of the documents also contain a large number of tables and diagrams, which is typical of technical regulatory material that relies heavily on specifications rather than continuous text.

This variation has a direct impact on how ingestion and retrieval are approached. Chunking and retrieval settings need to be flexible enough to handle short, highly targeted documents as well as longer standards that are split across multiple sections and parts.

6.3 Data Summary Visualisation

Figure 3 provides an overview of the dataset by showing the total number of pages for each document. The plot highlights the substantial variation in document length across the corpus, from relatively brief guidance documents to extensive technical standards.

In terms of overall content volume, the larger documents account for a significant proportion of the dataset, particularly the core AS/NZS standards and gas-related regulations. Because these documents contribute a large share of the total text, they require more careful chunking to preserve retrieval accuracy and ensure that citations remain precise.

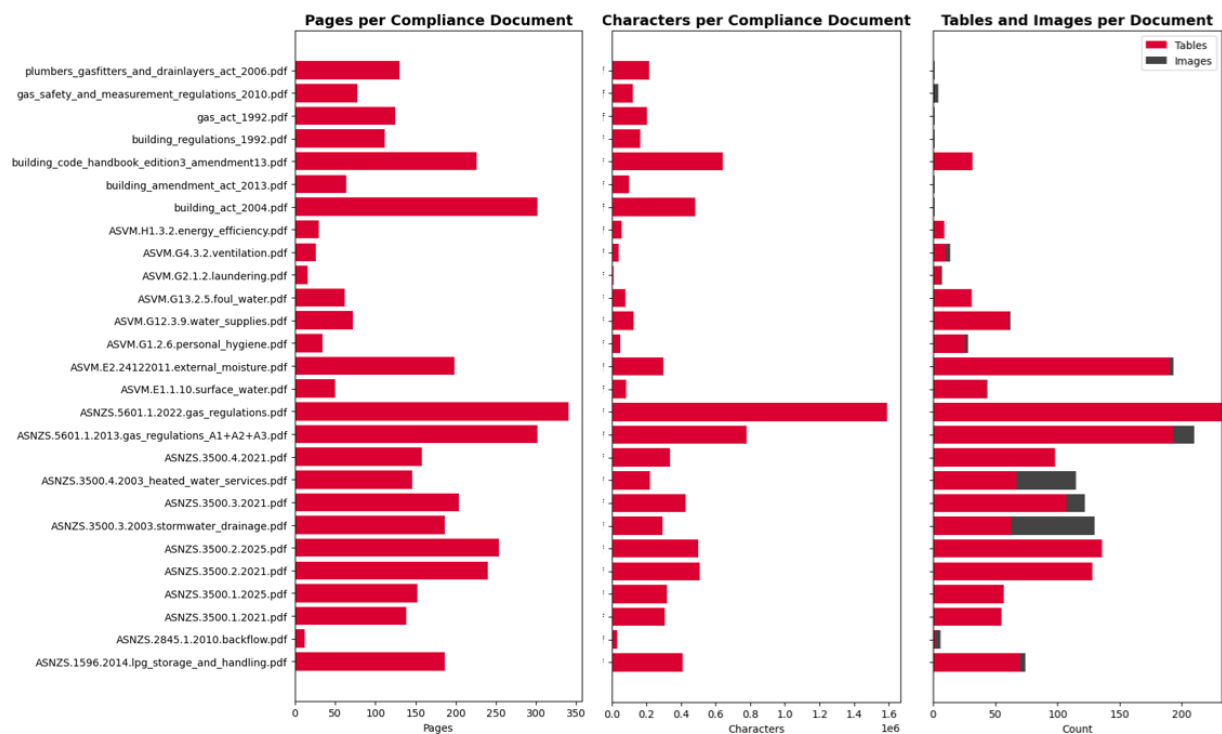


Figure 3 - Data Summary

6.4 Relevance to the RAG Approach

The size, complexity, and mixed nature of the dataset had a strong influence on the overall system design. Compliance questions are often tied to very specific clauses that can be buried deep within long documents, which meant the data needed to support page-level traceability and fine-grained chunking. The way the dataset is structured also led to the use of hybrid retrieval, since users tend to describe concepts in informal terms, while the source documents rely on formal and highly specific regulatory language.

Overall, the dataset provides a realistic and challenging foundation for evaluating retrieval-augmented generation in a compliance context, reflecting the conditions under which such a system would be used in practice.

7. Results

This section reports the observed performance of the prototype using the set of test questions captured in the interface screenshots. Results are presented for (i) retrieval coverage and ranking behaviour, (ii) answer quality and evidence grounding, (iii) ambiguity and missing-information handling, and (iv) end-to-end responsiveness (latency). The results are based on the system’s logged retrieval trace (dense + sparse + merge + rerank) and the final assistant outputs shown in the UI.

7.1 Question Coverage and Retrieval Performance

What was evaluated

Question coverage refers to whether the system can retrieve relevant evidence for a range of practical compliance questions (e.g., separation distances, gully traps, burial depth, venting, materials compatibility). Retrieval performance is assessed using the retrieval trace fields: dense hits, sparse hits, merged candidates, rerank candidates, and final top-k evidence.

Observed retrieval behaviour

Across the test set, the system consistently used a hybrid retrieval strategy:

- Dense (semantic) retrieval returned 20 candidates.
- Sparse (lexical) retrieval returned 20 candidates.
- Both lists were merged, producing 31–40 merged candidates depending on the query.
- A reranking stage then selected top_n = 5 evidence chunks for the LLM to answer from.

This consistent pattern indicates that the retrieval pipeline is stable and that the system is not dependent on a single retrieval method. Hybrid retrieval is particularly valuable in compliance documents because queries may include both technical terms (better matched by sparse search) and paraphrased natural language (better matched by dense embeddings).

Retrieval summary (from traces).

Table 1: Summary of Hybrid Retrieval Performance and Evidence Selection

Retrieval component	Typical observed result	Interpretation
---------------------	-------------------------	----------------

Dense hits	20	Semantic similarity search provides broad topical matches
Sparse hits	20	Keyword matching captures exact terms and clause language
Merged candidates	31–40	Candidate pool increases through hybrid merge
Rerank candidates	31–40	Reranker evaluates the merged set
Final top evidence	5	A compact evidence set is passed to the answer model

Edition governance signals

The trace also shows that some runs included evidence tagged as “Not yet referenced in Building Code” (2025 editions) alongside “Current” sources. This is important because it demonstrates the system can surface content from multiple editions while retaining metadata that supports correct interpretation (e.g., whether a standard is currently referenced in Acceptable Solutions).

Implication.

Overall, retrieval behaviour indicates the system can support a diverse set of plumbing/compliance queries and consistently return a manageable number of evidence chunks for answer generation. Stable retrieval counts do not automatically guarantee correct answers; correctness still depends on which chunks appear in the top 5 and whether those chunks contain the required detail (e.g., numeric limits).

7.2 Answer Quality and Evidence Grounding

What was evaluated

Answer quality is assessed by (1) whether the response addresses the user’s question clearly, (2) whether the answer includes relevant technical specifics (e.g., distances, test duration, required methods), and (3) whether claims are grounded in retrieved evidence displayed in the UI.

Observed strengths

The strongest outputs occurred when retrieved evidence contained explicit measurable requirements, such as:

- Separation distance rules (numeric distances and conditions),
- Testing requirements before sign-off (test type, timing, and “before backfill” constraint),
- Documentation requirements (product conformance and testing/certification evidence).

In these cases, answers were both actionable and strongly supported by evidence blocks and citations.

Observed limitations

Weaker answers occurred when:

- The question required a specific numeric requirement, but the retrieved pages did not include that number (e.g., minimum clearances).
- The question was broad, requiring a more constrained scope to answer precisely (e.g., “What materials can or cannot be used together?”).
- The retrieved evidence was general context rather than the exact clause or table row needed to answer definitively.

Table 2: Observed Answer Quality Across Different Question Types

Question type (example)	Observed quality	Explanation
Numeric rules (separation, testing)	Strong	Evidence contained direct numeric constraints, enabling grounded answers
Installation guidance (gully traps, venting)	Moderate–Strong	Multiple requirements supported, but some items depend on full clause/table context
Broad compliance checks (meet AS/NZS 3500)	Moderate	Answer becomes checklist-like; evidence supports general compliance but not a single definitive rule
Missing detail in retrieved pages (hot water cylinders, minimum clearances)	Weak but safe	The system stated that specific requirements were not present in the provided context (avoids hallucination)

Implication

Evidence-grounded answering is working as intended when relevant clauses/tables are retrieved. The main risk to answer quality is retrieval bottleneck: if the precise clause/table is not in the top-k evidence, the model cannot reliably produce a correct, specific answer.

7.3 Handling of Ambiguity and Missing Information

What was evaluated.

This subsection evaluates the system’s behaviour when (1) the user query is ambiguous or under-specified, or (2) the retrieved evidence does not include enough information to answer.

Ambiguity handling (broad queries)

Some questions inherently have multiple valid interpretations. For example:

- “*What materials can or cannot be used together?*” could mean dissimilar metals in contact, plastics-to-metals, compatibility in wet zones, coastal exposure, or potable water constraints.

In such cases, the system tended to provide general compatibility rules when available but did not always prompt for clarification. From a compliance perspective, missing context (environment, location, service type, materials in contact) can change the correct recommendation.

Missing-information handling (retrieval gaps)

In at least two cases, the system explicitly indicated that the retrieved sources did not contain the requested level of detail (e.g., hot water cylinder installation specifics; minimum clearances). This behaviour is desirable because it prevents unsupported or fabricated requirements.

Table 3: Observed System Behaviour Under Different Retrieval Scenarios

Scenario	System behaviour	Assessment
Query ambiguous	Provided general guidance; clarification not always requested	Acceptable, but clarification would improve precision
Evidence missing	Explicitly stated missing detail in context	Strong (reduces hallucination risk)
Mixed editions retrieved	Metadata flagged non-referenced 2025 material	Strong (supports governance and transparency)

Implication

The system demonstrates appropriate “safety” behaviour by acknowledging missing evidence. For ambiguous queries, adding a short clarification step would improve compliance usefulness without significantly increasing latency.

7.4 System Responsiveness

What was evaluated

Responsiveness is measured using the latency breakdown displayed in the trace: retrieval trace time, hybrid search (including rerank), answer generation time, and end-to-end totals.

Observed latency pattern

- Retrieval stages (trace + hybrid search) were relatively stable across questions.
- The largest variance came from answer generation, which increased for longer, multi-point responses.

Table 4: Latency Breakdown Across RAG Pipeline Components

Latency component	Observed range	Interpretation
Trace (lexical + semantic + merge)	0.79s – 4.15s	Retrieval setup and merging overhead
Hybrid search (incl. rerank)	~1.93s – 3.22s	Reranking adds cost but remains consistent
Answer generation	~6.76s – 21.77s	Main driver of slow runs (longer responses)
Pipeline total (hybrid + answer)	9.25s – 23.91s	End-to-end model work excluding trace
Cell total (trace + pipeline)	13.25s – 27.87s	Overall time a user experiences

Implication

The prototype is usable for interactive Q&A, but responsiveness is primarily constrained by generation time. In future iterations, response-time consistency could be improved by (i) limiting response verbosity, (ii) tighter evidence summarisation, or (iii) streaming and early stopping strategies to reduce perceived wait time.

8. Outcomes

8.1 Primary Project Outcomes

The key outcome of this project is the development of a functional proof-of-concept Compliance AI Assistant that enables evidence-based access to New Zealand building and plumbing standards. The system shows that retrieval-augmented generation, RAG, can be applied effectively to complex regulatory document collections, allowing users to ask questions in natural language and receive structured responses supported by citations. The assistant is able to

consistently retrieve relevant material from official standards and present answers with direct links to the exact page in the original document, which supports both transparency and verification.

This outcome directly responds to the original problem of slow, manual, and often inconsistent compliance searches, by delivering a practical decision-support tool rather than a system that is only experimental in nature.

8.2 Technical Outcomes

From a technical perspective, the project achieved several important outcomes:

- A hybrid retrieval pipeline combining dense semantic search with sparse lexical search was implemented successfully and shown to be reliable across a wide range of real-world compliance questions.
- Inference-based reranking using bge-reranker-v2-m3 improved the relevance of retrieved evidence, which made it possible for the system to operate with a relatively small but high-quality context window limited to the top-5 chunks
- Page-level traceability was preserved end-to-end, enabling precise citation links in the form `<s3-link>#page=<page_number>`.
- The system architecture also enabled real-time interaction through WebSockets, with responses streamed incrementally to improve the overall experience for users.
- A feedback capture mechanism was integrated, enabling users to rate responses and provide qualitative comments that can be used for future improvement.

Taken together, these outcomes show that the system is not only functional, but also designed in a way that supports auditability, scalability, and ongoing refinement.

8.3 Evaluation Outcomes

Evaluation using representative compliance questions provided by Foleys showed that:

- Overall, the system was able to handle many different types of questions and usually retrieved relevant evidence across plumbing and drainage topics, such as separation distances, testing requirements, material compatibility, and documentation requirements.
- Answers tended to be strongest in cases where the retrieved evidence contained clear numeric limits or step-by-step procedural rules, as this gave the model enough structure to produce responses that were specific and practical.

- Where the retrieved context did not include the required information, the system did not attempt to invent an answer and instead explicitly signalled that the available evidence was incomplete or insufficient.
- Retrieval traces and metadata confirmed that edition awareness (current vs not-yet-referenced standards) was preserved and surfaced appropriately edition awareness.

Overall, the evaluation indicates that the assistant behaves as a safe, evidence-grounded compliance support tool, rather than an over-confident generative system.

8.4 Practical Outcomes for Foleys

From an industry point of view, the project highlights several practical benefits for Foleys.:

- It enables faster access to authoritative information during quoting, planning, and on-site decision-making.
- It improves consistency and transparency, since every response is directly linked back to official source documentation.
- It also outlines a clear path for expanding the prototype into an internal tool, with wider document coverage and more formal validation workflows.

Although the system is not designed to replace professional judgement or formal regulatory approval processes, it provides a solid foundation for improving efficiency in everyday compliance-related work.

8.5 Limitations and Scope of Outcomes

It is important to note the boundaries of the outcomes achieved:

- The evaluation was conducted on a limited but representative question set, rather than large-scale quantitative benchmarking.
- Answer correctness remains dependent on successful retrieval of the relevant clause or table.
- Some regulatory questions require contextual details (environment, location, service type) that were not always available from the user query alone.

These limitations are appropriate for proof-of-concept and are addressed further in the Future Work section.

9. Conclusion

This project developed and evaluated a proof-of-concept Compliance AI Assistant for Foleys to support faster access to New Zealand plumbing and building compliance information. The main contribution was the data science work needed to make a large regulatory corpus usable: 27 authoritative documents were processed into chunked, metadata-rich records and indexed to support retrieval-augmented generation with traceable citations.

The results show that hybrid retrieval is practical for compliance workflows where users ask questions in everyday language but standards are written in technical terms. Across the evaluated questions, the pipeline consistently combined dense and sparse retrieval, merged candidates, reranked them, and used a small top-k evidence set to generate structured answers. A key strength is transparency: responses are grounded in official text and include page-level links so users can verify claims directly in the source PDFs. The system also behaved appropriately when evidence was missing, stating limitations rather than guessing.

Limitations remain. Answer quality is still constrained by the retrieval bottleneck, and some queries require additional situational detail to be answered precisely. Latency was acceptable for a proof-of-concept, but generation time was the main contributor to slower responses. The choice of the gpt-4.1-mini model reflects a deliberate design trade-off: prioritising structured reasoning, prompt adherence, and cost-efficient deployment over maximal generative capability. In this context, the language model's role is to faithfully synthesise retrieved evidence rather than to reason independently, making this model appropriate for an evidence-grounded compliance assistant. Overall, the project demonstrates that an evidence-grounded RAG assistant can reduce manual document searching and improve traceability, providing a solid foundation for future refinement in version governance, ambiguity handling, and evaluation at a larger scale.

10. Future Work

Future work should look at expanding both the scope of the system and how it is used in real settings. At present, the system only indexes extracted text, which means diagrams and other image-based material commonly found in standards are not captured. Adding support for image-aware retrieval, for example allowing users to upload an image and locate related clauses or figures, would improve coverage. Performance is another area that would need attention, especially reducing end-to-end latency to around five seconds so the system can be used

realistically on site. One possible way to address this is by placing tighter limits on generation and introducing caching in parts of the pipeline where it would have the most impact. From a usability point of view, the system would also benefit from letting users upload PDFs directly through the interface, instead of depending on manual ingestion steps. Over time, adding duplicate document detection would make it easier to manage sources properly and help prevent declines in retrieval quality as more documents are added.

References

- Adobe. (2007, April). *Parameters for Opening PDF Files*. From PDF Objects: https://pdfobject.com/pdf/pdf_open_parameters_acro8.pdf
- Arunabh, B., & Heriberto, C. (2024). Systematic Analysis of Retrieval-Augmented Generation-Based LLMs for Medical Chatbot Applications. *Machine Learning & Knowledge Extraction*, 2356-2356.
- Maryamah, M., Muhammad, M. I., Edric, B. T., Netri, A. R., Mohammad, G., & Indra, K. R. (2024). Chatbots in Academia: A Retrieval-Augmented . *2024 16th International Conference on Knowledge and Smart Technology (KST)* (pp. 259-264). Krabi, Thailand: IEEE. From PyMuPDF Documentaion: <https://pymupdf.readthedocs.io/en/latest/recipes-text.html>
- Ministry of Business, Innovation & Employment. (2014, February 14). *New Zealand Building Code Handbook*. From Building Gov NZ: <https://www.building.govt.nz/assets/Uploads/building-code-compliance/handbooks/building-code-handbook/building-code-handbook-3rd-edition-amendment-13.pdf>
- Ministry of Business, Innovation & Employment. (2016, March 21). *Alternative solutions for compliance with the Building Code*. From Building Gov NZ: <https://www.building.govt.nz/building-code-compliance/how-the-building-code-works/different-ways-to-comply/alternative-solutions>
- Ministry of Business, Innovation & Employment. (2021, May 28). *AS/NZS 3500.1:2021 Plumbing and drainage Part 1: Water services*. From Building CodeHub Nz: <https://codehub.building.govt.nz/resources/asnzs-3500-12021>
- Ministry of Business, Innovation & Employment. (2026, January 28). *Acceptable solutions and verification methods*. From Building Gov NZ:

<https://www.building.govt.nz/building-code-compliance/how-the-building-code-works/different-ways-to-comply/acceptable-solutions-and-verification-methods>

- MongoDB. (n.d.). *MongoDB's Flexible Schema: Unpacking the "Schemaless Database"*. From MongoDB: <https://www.mongodb.com/resources/basics/unstructured-data/schemaless>
- OpenAI. (2022, December 16). *How to count tokens with Tiktoken*. From OpenAI Developers: https://developers.openai.com/cookbook/examples/how_to_count_tokens_with_tiktoken
- OpenAI. (n.d.). *GPT-4.1 mini*. From OpenAI Platform: <https://platform.openai.com/docs/models/gpt-4.1-mini>
- OpenAI. (n.d.). *tiktoken*. From GitHub, Inc: <https://github.com/openai/tiktoken>
- Pinecone Systems, Inc. (n.d.). *Hybrid search*. From Pinecone Systems, Inc: <https://docs.pinecone.io/guides/search/hybrid-search>
- Pinecone Systems, Inc. (n.d.). *Models: bge-reranker-v2-m3*. From Pinecone Systems, Inc: <https://docs.pinecone.io/models/bge-reranker-v2-m3>
- Pinecone Systems, Inc. (n.d.). *Models: pinecone-sparse-english-v0*. From Pinecone Systems, Inc.
- PyMuPDF. (n.d.). *Text extraction recipes / tutorial*. From PyMuPDF Documentation: <https://pymupdf.readthedocs.io/en/latest/recipes-text.html>
- STANDARDS AUSTRALIA. (2021, May). *Standards Catalogue - AS/NZS 3500.3:2021*. From STANDARDS AUSTRALIA: <https://www.standards.org.au/standards-catalogue/standard-details?designation=as-nzs-3500-3-2021>

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Appendices

Appendix A: Links

Deployed application: <https://foleys-ai-assistance-app-production.up.railway.app/>

Email: nirosigaa@mail.com

Password: 123456

Appendix B: Database Schema

Postgres Schema

Users table

This table stores authenticated user information and is used during login, authorization checks, and access control.

- id (UUID) – unique user identifier
- email (TEXT) – user email address (unique)
- password_hash (TEXT) – securely hashed password
- name (TEXT) – user display name
- role (TEXT) – user role (for example USER or ADMIN)
- company (TEXT, nullable) – organisation affiliation
- is_active (BOOLEAN) – account status
- created_at (TIMESTAMP) – account creation time
- updated_at (TIMESTAMP) – last update time

Conversation analytics table

This table records structured usage metrics for each assistant response. It enables analysis of system performance, cost, and usage patterns.

- id (UUID) – analytics record identifier
- user_id (UUID) – reference to users table
- user_name (TEXT) – user name at time of interaction
- company (TEXT, nullable) – company name
- conversation_id (TEXT) – MongoDB conversation reference
- message_id (TEXT, nullable) – MongoDB message reference
- model (TEXT) – language model used
- prompt_tokens (INTEGER) – number of prompt tokens

- completion_tokens (INTEGER) – number of completion tokens
- total_tokens (INTEGER) – total tokens used
- latency_ms (INTEGER) – response latency in milliseconds
- created_at (TIMESTAMP) – record creation time

Feedback analytics table

This table stores structured feedback data for reporting and dashboard visualisation.

- id (UUID) – feedback record identifier
- user_id (UUID) – reference to users table
- user_name (TEXT) – user name
- conversation_id (TEXT) – conversation reference
- message_id (TEXT) – message reference
- rating (TEXT) – feedback rating (positive or negative)
- category (TEXT) – feedback category
- feedback_text (TEXT, nullable) – optional user comment
- created_at (TIMESTAMP) – feedback submission time

MongoDB Schema

Conversations collection

Each document represents a single chat session.

- _id (ObjectId) – conversation identifier
- userId (ObjectId or STRING) – user reference
- title (STRING) – conversation title
- summary (STRING, nullable) – optional summary
- messageCount (INTEGER) – number of messages
- createdAt (DATETIME) – creation timestamp
- updatedAt (DATETIME) – last update timestamp
- deletedAt (DATETIME, nullable) – soft delete marker

Messages collection

Each document represents a single message within a conversation.

- _id (ObjectId) – message identifier
- conversationId (ObjectId or STRING) – conversation reference

- userId (ObjectId or STRING) – user reference
- role (STRING) – message role (user or assistant)
- content (STRING) – message text
- sequence (INTEGER) – ordering within conversation
- timestamp (DATETIME) – message time

Message metadata (embedded document)

Stored only for assistant messages to support evaluation and analytics.

- model (STRING) – language model used
- promptTokens (INTEGER, nullable) – prompt token count
- completionTokens (INTEGER, nullable) – completion token count
- latencyMs (INTEGER, nullable) – response latency
- sources (ARRAY) – retrieved source metadata

Message feedback collection

Stores user feedback linked to individual assistant messages.

- _id (ObjectId) – feedback identifier
- messageId (ObjectId or STRING) – message reference
- conversationId (ObjectId or STRING) – conversation reference
- userId (ObjectId or STRING) – user reference
- rating (STRING) – positive or negative
- comment (STRING, nullable) – optional feedback text
- categories (ARRAY of STRING) – selected feedback categories
- submittedAt (DATETIME) – feedback submission time
- createdAt (DATETIME) – record creation time
- updatedAt (DATETIME) – last update time

Pinecone Chunk record metadata (stored in both dense and sparse indexes)

- id (STRING) – unique chunk identifier
- chunk_text (STRING) – extracted text content
- doc_title (STRING) – document title
- filename (STRING) – source PDF filename

- file_link (STRING) – public S3 URL
- page_number (INTEGER) – source page number
- standard_type (STRING) – standard category (e.g. AS/NZS, NZBC)
- edition_year (INTEGER) – document edition year
- currency_status (STRING) – current or superseded

Appendix C: User Interface Design

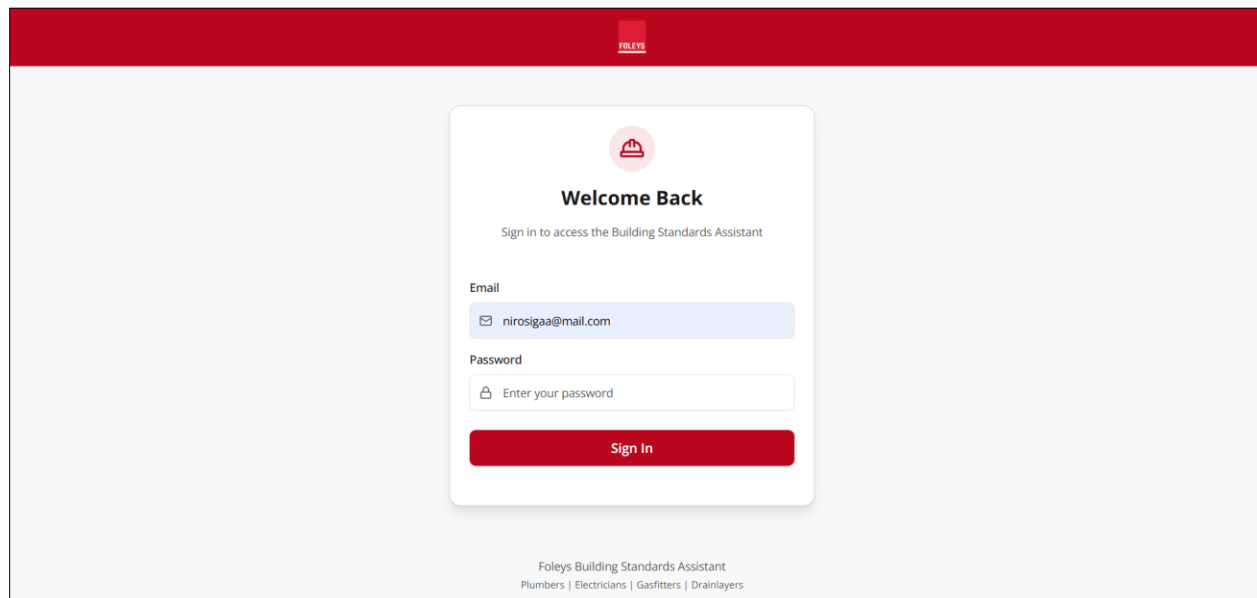


Figure 4 - Login page

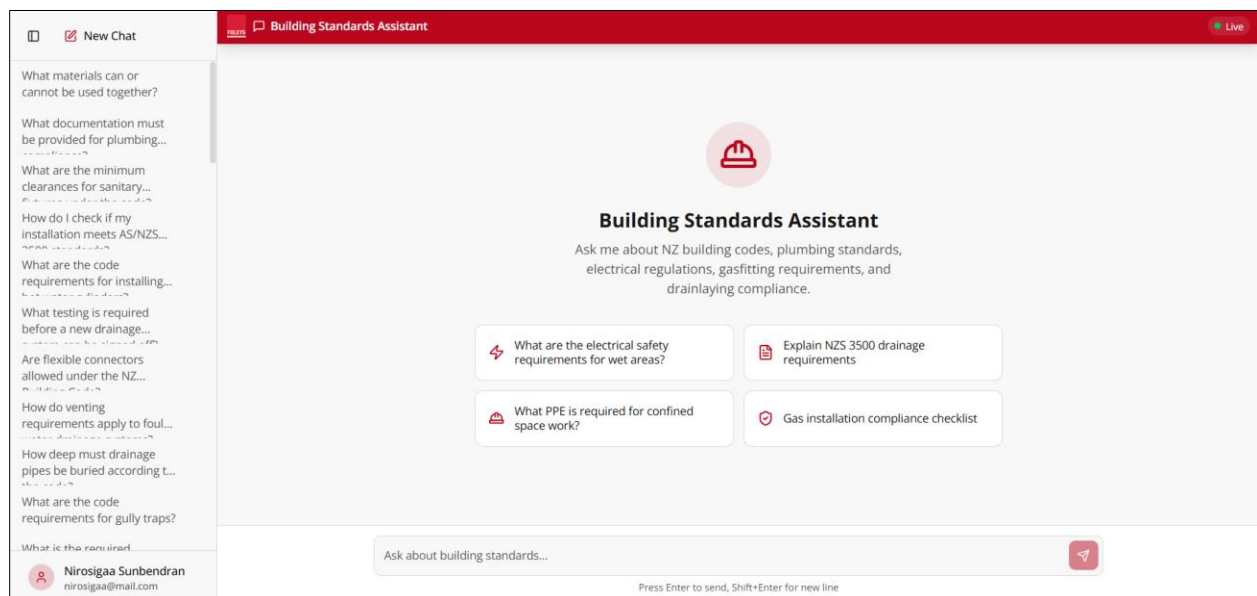


Figure 5 - Chat page

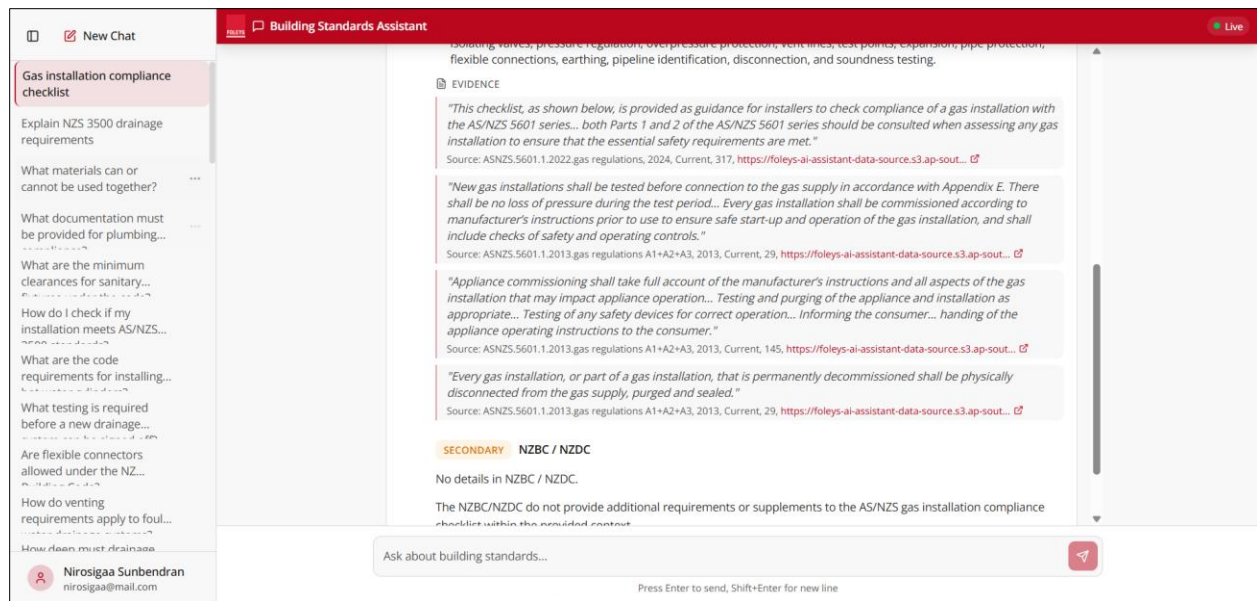


Figure 6 - Chat response

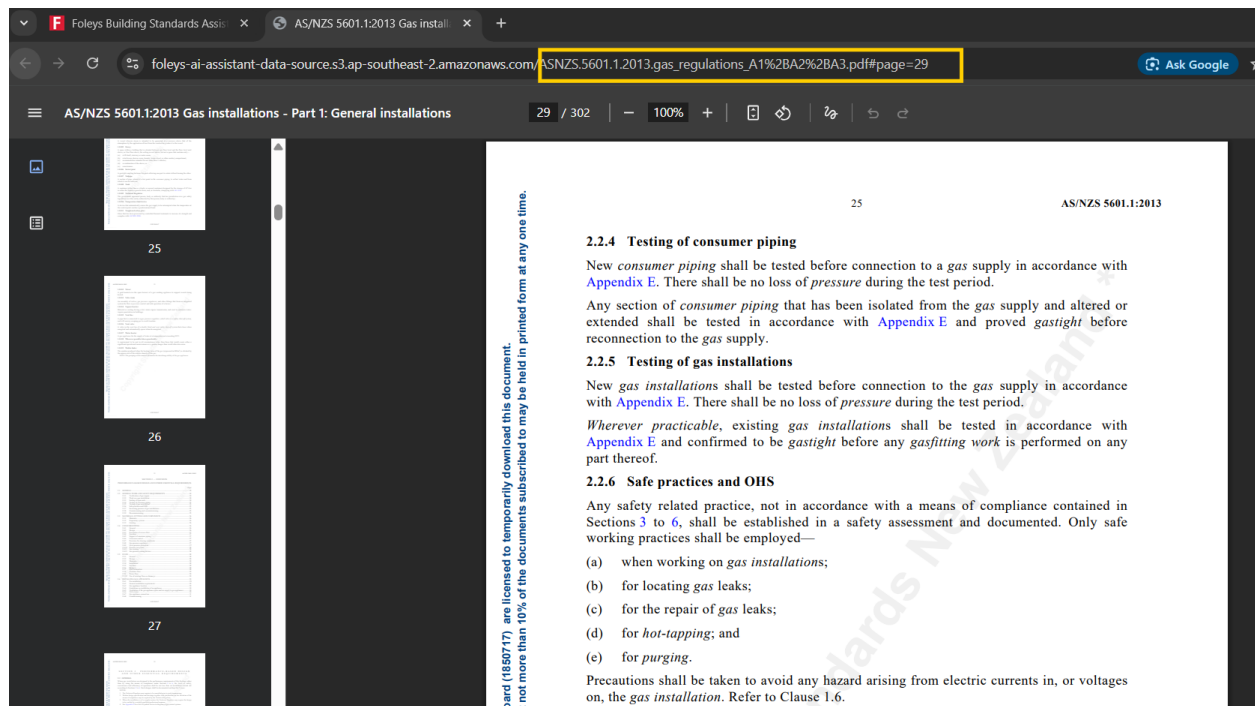


Figure 7 - PDF Navigation

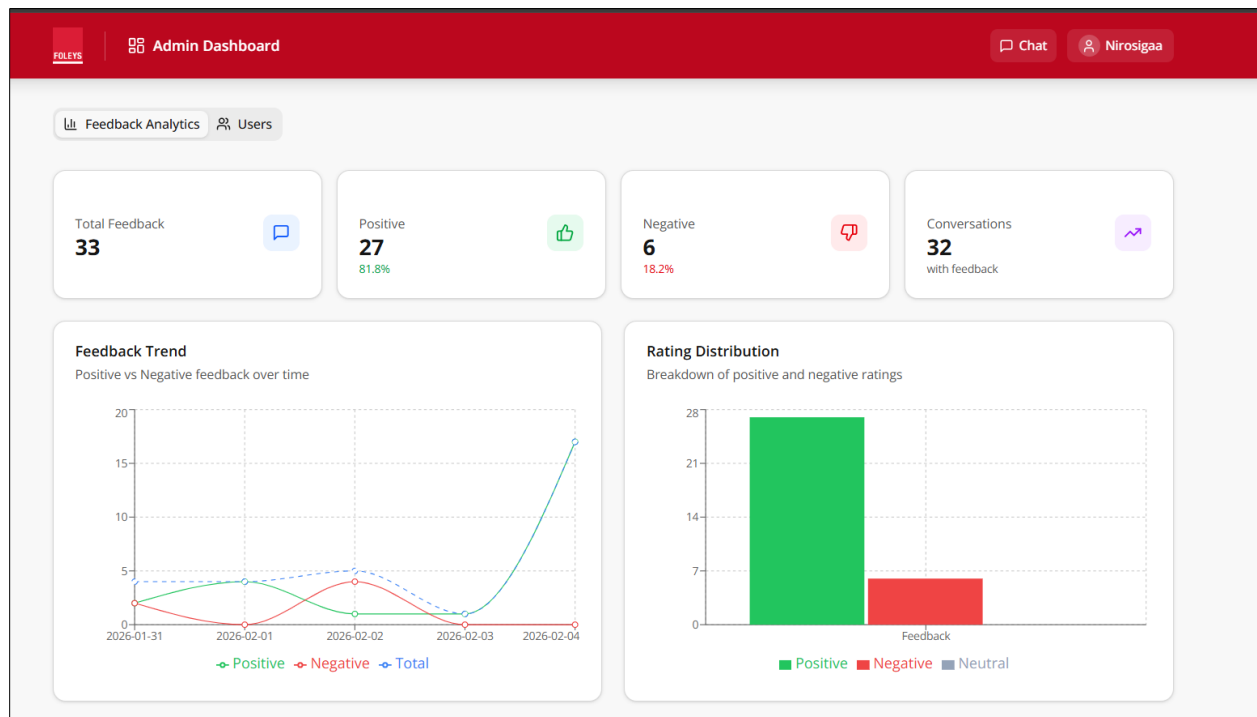


Figure 8 - Admin dashboard(Analytics)

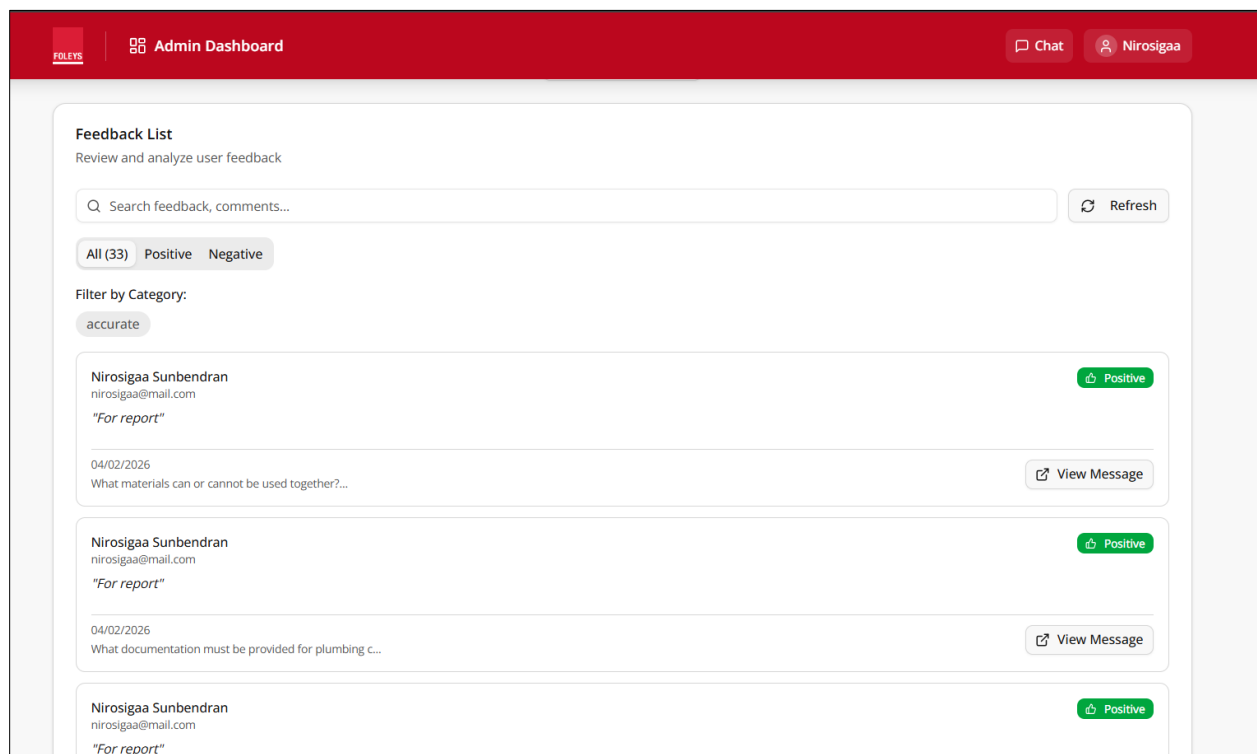


Figure 9 - Admin dashboard(Feedback List)

Appendix D: AWS S3 Bucket

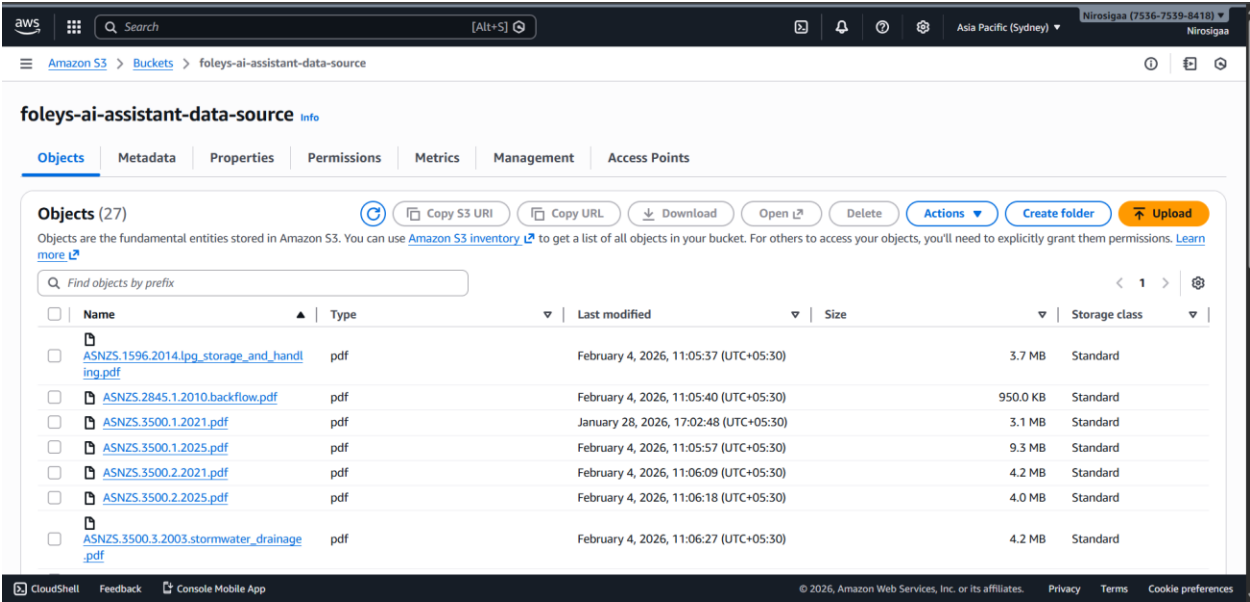


Figure 11 - S3 Bucket for PDF Storage

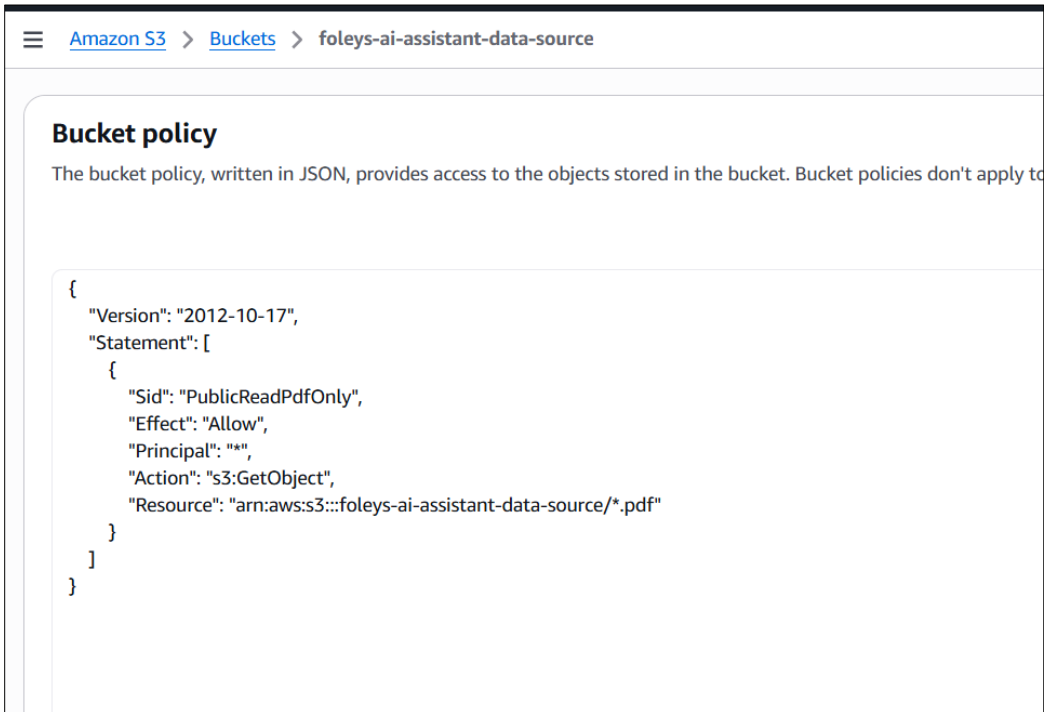


Figure 12 - S3 Bucket Public Policy

The screenshot shows a PostgreSQL query editor with three SQL queries in the top pane:

```
select * from public.users
select * from public.conversation_analytics
select * from public.feedback_analytics
```

The bottom pane displays the 'users' table with the following data:

	id	email	password_hash	first_name	last_name	role	company	created_at
1	c48c4483-3e4a-4a5d-a3b2-1277066d18de	jhon@mail.com	\$pbkdf2-sha256\$290005\$FqD5FEaA18CfWbsI2aw\$XbuRwuBFqSZob5Yelo/AiOIP	Jhon	Andrew	admin	folyes	2026-01-27 20:50:05.456
2	d2e005fe-5ce9-42f1-b5c8-76f82db702ca	nirosigaa@mail.com	\$pbkdf2-sha256\$290005\$wffUwpBaG0ZayVMgYgvASdtkATx2Ah/qPcAtD/W9y3EZg	Nirosigaa	Sunbendran	admin	folyes	2026-01-27 20:28:25.089

Figure 15 - users postgres table

Appendix F: MongoDB Collection

The screenshot shows the MongoDB Compass interface for the 'folyes-ai-chat.conversations' collection. The top right corner indicates 103 documents and 3 indexes. The 'Documents' tab is selected, showing a list of documents. The first document is expanded, showing the following JSON structure:

```
{
  "_id": ObjectId('697a345f5dc18c648abe2cd2'),
  "userId": "d2e005fe-5ce9-42f1-b5c8-76f82db702ca",
  "title": "Explain about stormwater drainage",
  "summary": null,
  "messageCount": 4,
  "lastMessagePreview": "## AS/NZS (Primary)
    - Use and storage of LP Gas cylinders indoors sho...",
  "createdAt": 2026-01-28T16:07:59.251+00:00,
  "updatedAt": 2026-01-28T16:11:37.142+00:00,
  "deletedAt": null
}
```

The second document is also expanded, showing the following JSON structure:

```
{
  "_id": ObjectId('697a35e6934cc3f77d8ad342'),
  "userId": "d2e005fe-5ce9-42f1-b5c8-76f82db702ca",
  "title": "how to use lp gas",
  "summary": null,
  "messageCount": 4,
  "lastMessagePreview": "## AS/NZS (Primary)
    - Stormwater drainage systems associated with LP ...",
  "createdAt": 2026-01-28T16:14:30.641+00:00,
  "updatedAt": 2026-01-28T16:22:18.775+00:00,
  "deletedAt": null
}
```

Figure 16 - conversations mongo collection

foleyes-ai-chat.messages

Documents Aggregations Schema Explain Plan Indexes Validation

Filter Type a query: { field: 'value' }

ADD DATA EXPORT COLLECTION

```

_id: ObjectId('697a345f5dc18c648abe2cd3')
conversationId: ObjectId('697a345f5dc18c648abe2cd2')
userId: "d2e005fe-5ce9-42f1-b5c8-76f82db702ca"
role: "user"
content: "Explain about stormwater drainage"
timestamp: 2026-01-28T16:07:59.393+00:00
sequence: 1

_id: ObjectId('697a34795dc18c648abe2cd4')
conversationId: ObjectId('697a345f5dc18c648abe2cd2')
userId: "d2e005fe-5ce9-42f1-b5c8-76f82db702ca"
role: "assistant"
content: "## AS/NZS (Primary)

- Stormwater drainage systems associated with LP ..."
timestamp: 2026-01-28T16:08:25.344+00:00
sequence: 2
▼ metadata: Object
  model: "gpt-4.1-mini"
  latencyMs: 25672
▼ sources: Array
  ▼ 0: Object
    title: "ASNZS.1596.2014.lpg storage and handling"
    filename: "ASNZS.1596.2014.lpg_storage_and_handling.pdf"
    file_link: "file:///D:/mis/foleyes-ai-assistance/foleyes-ai-assistance-service/PDFs/..."
    page_number: 163
    standard_type: "General"
    text_preview: "159 AS/NZS 1596:2014
    *
    APPENDIX K
    EMERGENCY PLANNING d
    n
    (Informative)..."
  
```

Figure 17 - messages mongo collection

foleyes-ai-chat.message_feedback

Documents Aggregations Schema Explain Plan Indexes Validation

Filter Type a query: { field: 'value' }

ADD DATA EXPORT COLLECTION

```

-----
conversationId: ObjectId('697c0600be09bdc3fc9fd54')
userId: "d2e005fe-5ce9-42f1-b5c8-76f82db702ca"
rating: "positive"
comment: null
▼ categories: Array
  0: "accurate"
submittedAt: 2026-01-31T05:36:44.906+00:00
createdAt: 2026-01-31T05:09:21.420+00:00
updatedAt: 2026-01-31T05:36:44.906+00:00

_id: ObjectId('697d98b24de99604fddc5346')
messageId: ObjectId('697d98984de99604fddc5345')
conversationId: ObjectId('697d987d4de99604fddc5343')
userId: "d2e005fe-5ce9-42f1-b5c8-76f82db702ca"
rating: "positive"
comment: "Test this one"
▼ categories: Array
  submittedAt: 2026-01-31T12:39:50.198+00:00
  createdAt: 2026-01-31T05:52:50.024+00:00
  updatedAt: 2026-01-31T12:39:50.198+00:00
  
```

Figure 18 - message_feedback mongo collection

Appendix G: LLM Prompt

```
system_prompt = """
You are a compliance assistant for New Zealand building and plumbing standards.
You must answer ONLY using the provided context extracted from official documents.

Priorities:
1) Accuracy and traceability are more important than completeness.
2) Do not guess, do not assume, and do not add external knowledge.
3) If the context is insufficient or ambiguous, ask 1-2 clarification questions in bold and stop.

Edition and superseding handling:
- Prefer sources marked as current (is_current = true).
- If you use any non-current source (superseded or not yet referenced), you MUST label it clearly and briefly explain why it appears.
- If multiple editions of the same standard appear in the context, base the main requirements on the current edition unless the user explicitly asks for a comparison.

Currency label rules (use metadata from context):
- If is_current is true: currency_status = "Current"
- If note indicates "Superseded version": currency_status = "Superseded"
- If note indicates "Not yet referenced in Building Code": currency_status = "Not yet referenced"

Citations and links:
- Every claim must be supported by at least one exact quote copied from the context.
- For every evidence quote, you MUST output a single clickable URL by combining file_link and page_number exactly like this:
link = file_link + "#page=" + page_number
- The final evidence format must be: (filename, year, currency_status, page_number, link)
- Do not include citations like [Source 1] or references to the retrieval process.
- Do not output the raw words "file_link#page=page_number". Always output the fully combined link.

Writing style: You, yesterday + prompt enhancement
- Use clear, concise language aimed at trade and compliance users.
- Keep answers practical and action-oriented.
"""
```

Figure 19 - LLM System Prompt

```
user_prompt = f"""
Return the answer in the exact structure below and follow all rules.

## AS/NZS (Primary)
Write 3-5 sentences explaining what the bullet points cover.
Then provide bullet points of requirements based ONLY on AS/NZS sources in the context.

Evidence quotes:
- "<exact quote>" (filename, year, currency_status, page_number, <combined_link>)
- "<exact quote>" (filename, year, currency_status, page_number, <combined_link>)

Missing details include: <brief list or 'none'>.

## NZBC / NZDC (Secondary)
Provide bullet points of requirements based ONLY on NZBC/NZDC sources in the context.

Evidence quotes:
- "<exact quote>" (filename, year, currency_status, page_number, <combined_link>)
- "<exact quote>" (filename, year, currency_status, page_number, <combined_link>)

Write 1 sentence explaining how NZBC/NZDC supplements the AS/NZS requirements (based only on the context).

Rules:
- If there is no NZBC/NZDC support in the context, write exactly: "No details in NZBC / NZDC."
- If there is no AS/NZS support in the context, write exactly: "No details in AS/NZS."
- Always combine links using: file_link + "#page=" + page_number.
- The combined link must start with https:// and must end with #page=<number>.
- Do not invent standards, clause numbers, values, or requirements.
- Use exact wording from the context for quotes (copy-paste).
- Do not include [Source N] citations or mention chunk IDs.
- Keep the response concise.
- If the context is not sufficient to answer confidently, ask 1-2 clarification questions in bold and stop.

Context:
{context}

Question:
{question}
"""
```

Figure 20 - LLM User Prompt